Take-home midterm PHYS 508

Prof . Bunker Fall 2025

You are free to use books, internet, Mathematica etc, whatever, ***but you must do the problems by yourself**,* preferably using your notes and what is between your ears.
Don't use ChatGPT or other LLMs, I don't want to have to grade nonsense, and I want you to learn. Be prepared to defend every thing you write.

Problems are due Wednesday Oct 22 (or before), preferably electronically (scans or photo), although paper is OK.
If you have questions or are stuck don't hesitate to email me at bunker@iit.edu, or ask in class.

have fun!
gb

Problem 1

A body of mass *m* is in a potential U(x,y)= $\frac{\epsilon}{2}\left(\left|\frac{x}{a}\right|+\left|\frac{y}{b}\right|\right)$ where ϵ, *a*, and *b* are positive constants.
This is a homogeneous potential (considering only positive scale factors α) so "mechanical similarity" applies as described in lecture.
a) what is the degree of homogeneity *k* of the potential?
b) if the total energy is $\mathcal{E}$, what are the long time average values <T> and <U> of the kinetic and potential energy ?
c) If the energy $\mathcal{E}$ in the system is increased by a factor ϕ (i.e. $\mathcal{E} \rightarrow \phi^* \mathcal{E}$), determine the factors by which the following quantities are changed:

i) size of orbit
ii) orbital period
iii) velocity
iv) angular momentum
v) action

Problem 2

a) Write down the Lagrangian for the system described in problem 1.
b) Show that the problem decomposes into two separate sub-problems, one in x, and one in y.
c) Although the absolute value function is not differentiable, a good approximation to it is the function

abs(x)=*w**ln(2 cosh(x/w))

where *w* is a very small positive number (w is a parameter describing the width of the region over which there is a significant change in slope). This abs(x) function *is* differentiable.
Pick a value of w and plot this function in x from -10w to 10w using a suitable program such as Mathematica.

d) If the potential is given above is used with the given abs(x) function (as one might in a numerical simulation), what are the two equations of motion (in *x* and *y*)?
You are *not* asked to solve them here, but the problem is now suitable for use in a differential equation solver (say) NDSolve.

problem 3

A body of mass m is in a potential

$$U(x) = \frac{\epsilon}{2}\left(\left(\frac{x}{a}\right)^2 + \left(\frac{a}{x}\right)^2\right)$$

where ϵ and a are positive constants.

i) Find the turning points

ii) calculate the period of **large** oscillations as a function of energy $\mathcal{E}$ in the system. Completing the square may be useful in the integral

iii) For very large amplitudes compare your results to the "one–sided potential" $U(x) = \frac{\epsilon}{2}\left(\frac{x}{a}\right)^2$, $x > 0$. Does your result make sense, physically?

problem 4

A particle of charge *q* and mass *m* is constrained to move along the z-axis. It has position z. A uniform ring of total charge Q (of opposite sign to *q*) and radius *R* is centered at the origin and is constrained to be in the x-y plane.
i) if a particle starts at z=0 with an initial z velocity v0, what is the minimum velocity needed for the particle to escape to z→∞?
ii) Find the frequency of *small* oscillations in the potential.
iii) write an expression (integral) for the frequency of *large* oscillations, parameterized by the energy $\mathcal{E}$. You don’t have to do the integral, it’s no fun.
iv) If the ring were allowed to move in the ± z-direction, and it had a mass *M*, how would the frequency of small oscillations be changed ?(give an *exact* expression)

problem 5

i) from the definition $\vec{L} = \vec{r} \times \vec{p}$ prove that $\frac{d\vec{L}}{dt} = 0$ for a central potential, i.e. $\vec{L}$ is conserved
ii) prove that the motion in a central potential is entirely in a plane (hint: assume it’s not and arrive at a contradiction).
iii) noting that $\vec{L}$ is perpendicular to the plane of motion, in what direction is the vector $\vec{F} \times \vec{L}$, where $\vec{F}$ is the (central) force?
iv) prove that in polar coordinates $\vec{F} \times \vec{L} = |\vec{F}|\,|\vec{L}|\,\hat{\phi}$ where $\hat{\phi}$ is the unit vector that points in the direction in which ϕ is increasing; $\hat{r}$ points in the direction in which r is increasing. hint: think of the geometry of the vectors, in particular, orthogonality.
v) recalling that $\hat{r} = (\cos(\phi), \sin(\phi))$; $\hat{\phi} = (-\sin(\phi), \cos(\phi))$, show that $\frac{d\hat{r}}{dt} = \hat{\phi}\,\frac{d\phi}{dt}$

vi) recalling that $|\vec{L}| = \mu\, r^2 \frac{d\phi}{dt}$ show that, for an inverse square force, $\vec{F} \times \vec{L}$ - *const* $\frac{d\hat{r}}{dt}$=0 for an appropriate choice of *const.*
For a potential U= -γ/r, what is that constant?
vii) show that $\vec{F} \times \vec{L} = \frac{d}{dt}(\vec{p} \times \vec{L})$, and therefore $\frac{d}{dt}(\vec{p} \times \vec{L}$ - const $\hat{r}) = 0$ using the constant *const* found above.
viii) conclude that $\vec{p} \times \vec{L}$ - const $\hat{r}$ is a conserved quantity for 1/r potential (inverse square law force).
Does a similar construction of a conserved quantity work for other central potentials? Why or why not?

Problem 6

Two masses m1 and m2 can move freely in the x-y plane (*both translating and rotating*) and they are hooked together by a spring of force constant k and zero natural length. The masses rotate around their center of mass with variable distance and rate of rotation.
Using plane polar coordinates (r, ϕ):
i) Write the Lagrangian
ii) identify any conserved quantities
iii) write the equations of motion. You are *not* asked to solve them.
They could be numerically integrated without difficulty.